

AI Safety Interventions: A Consolidated Report

Gunnar Zarncke (AE Studio)

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Abstract

This report consolidates catalogs of AI risk interventions: (1) the catalog of AI safety interventions from formal to governance approaches, (2) underexplored and neglected intervention areas with supporting references, and (3) newly identified gaps with little to no coverage in published literature. The synthesis orders interventions from hard science methods with provable guarantees through interpretability, alignment training, oversight, robustness, operational controls, provenance, psychology, brain-like approaches, and governance measures. The report aims to serve as a structured map of the intervention space and a guide for prioritizing neglected areas.

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1 Scope and Definitions

1.1 What Constitutes an AI Safety Intervention

For the purposes of this report, an **AI safety intervention** or **alignment risk reduction intervention** is defined as any technical, operational, or governance approach that aims to reduce the probability or severity of harmful outcomes from AI systems. This encompasses:

1. **Technical interventions** that modify AI system design, training, or inference to improve safety properties—including formal verification, interpretability methods, alignment training techniques, and robustness measures.
2. **Operational interventions** that establish processes, controls, or infrastructure around AI development and deployment—including oversight mechanisms, evaluation protocols, containment measures, and monitoring systems.
3. **Governance interventions** that create institutional frameworks, regulations, standards, or incentive structures to guide safe AI development—including policy frameworks, liability mechanisms, and coordination protocols.

1.2 Risk Categories Addressed

The interventions cataloged address multiple categories of AI risk:

- **Misalignment risks:** Systems pursuing objectives misaligned with human values or intentions
- **Capability risks:** Rapid capability gains outpacing safety measures or creating dual-use potential
- **Deception and manipulation risks:** Systems learning to deceive overseers or manipulate users
- **Robustness risks:** Failures under distribution shift, adversarial inputs, or edge cases
- **Systemic risks:** Emergent harms from widespread deployment or multi-agent interactions
- **Governance risks:** Inadequate oversight, coordination failures, or misuse by human actors
- **Existential risks:** Risks that are so severe that they threaten the survival of the human species or the planet.

1.3 Scope Boundaries

This report focuses specifically on interventions designed to reduce risks from advanced AI systems with general capabilities. It excludes:

- General cybersecurity measures not AI-specific

- Privacy protections unrelated to safety (though some overlap exists)
- Narrow AI safety (e.g., autonomous vehicle safety) unless generalizable
- Economic or social policy responses that don’t directly target AI risk reduction

The interventions span the full development lifecycle from research and training through deployment and governance, recognizing that effective risk reduction requires coordinated approaches across multiple stages and stakeholders.

2 Methodology

The methodology follows three steps:

1. **Source aggregation.** We drew on eight primary sources: the comprehensive AI alignment survey [Ji et al., 2023b], control measures overview [Greenblatt, 2023], alignment project ideas catalog [BlueDot Impact, 2023], the AI safety index report [Russell et al., 2025], the circuits research landscape [Lindsey et al., 2025], Anthropic’s foundational challenges framework [Ganguli et al., 2024], the technical AI governance research agenda [Reuel et al., 2024], and the governance extinction risk framework [Barnett and Scher, 2024], plus additional structured summaries, the AI Safety Map institutional overview [AI Safety Map Team, 2024], OpenAI Deep Research, and manual analyses.
2. **Synthesis.** We merged overlapping entries, expanded with prior work references (papers, Alignment Forum, LessWrong, governance docs), and ordered them across the continuum of human knowledge disciplines.
3. **Gap identification.** Areas lacking substantive publications or concrete proposals were identified as neglected and presented explicitly.

3 Foundational Theories of Agency and Decision-Making

Many alignment interventions—from shutdown safety and corrigibility, to interpretability and oversight—implicitly rely on assumptions about what it means for a system to be an *agent*. Without precise mathematical and conceptual foundations, interventions risk resting on fragile or ad-hoc assumptions. Below, existing research agendas are summarized that address these root questions.

3.1 Embedded Agency

Moving beyond the Cartesian boundary model to agents that exist within and interact with their environment [Demski and Garrabrant, 2018]. This foundational work addresses how agents can model themselves and reason under resource constraints, providing conceptual foundations for interventions that assume self-modeling capabilities. Critical for understanding how AI systems operate as embedded optimizers rather than external controllers. The PreDCA (Precursor Detection, Classification and Assistance) framework extends these concepts by proposing methods for AI systems to identify and assist user agents [Kosoy, 2022].

3.2 Decision Theory and Rational Choice

Foundations for rational choice under uncertainty, including causal vs. evidential decision theory and updateless decision theory [Hutter, 2005]. Functional Decision Theory (FDT) represents a newer approach where agents choose actions based on their decision function, aiming for better outcomes in scenarios involving multiple agents or copies [Yudkowsky and Soares, 2017]. These frameworks shape interventions in corrigibility and cooperation by providing formal models of how agents should reason about their own actions and their effects on the world. Relevant to shutdown safety and multi-agent coordination.

3.3 Optimization and Mesa-Optimization

Understanding when and how learned systems themselves become optimizers, with implications for deception and alignment faking. Mesa-optimizers emerge when a system optimized for one objective develops internal optimization processes that may pursue different goals. This research informs interpretability targets and oversight design by clarifying what internal structures to monitor.

3.4 Logical Induction

MIRI’s framework for reasoning under logical uncertainty provides a computable algorithm that assigns probabilities to logical statements and refines them over time [Garra-brant et al., 2016]. This addresses the challenge of reasoning when deductive resources are limited, offering foundations for AI systems to make informed decisions under logical uncertainty. Critical for developing oversight and corrigibility mechanisms when agents must reason about systems as complex as themselves.

3.5 Cartesian Frames and Finite Factored Sets

Mathematical formalization of agency and decision-making that provides tools for manipulating sets of actions and understanding how agents interact with their environments [Garra-brant et al., 2021]. This framework offers precise definitions for agency concepts crucial to alignment interventions, enabling formal analysis of agent-environment boundaries and decision-making processes. Finite Factored Sets extend this work by providing mathematical foundations for causality and temporal inference, offering a novel framework for understanding causal relationships [Garra-brant, 2020]. Together, these approaches provide tools for modeling complex dependencies and independencies in AI systems, supporting formal analysis of both agency and causal reasoning in safety-critical applications.

3.6 Infra-Bayesianism and Logical Uncertainty

Efforts to handle uncertainty in logical domains and imperfect models, addressing limitations of standard Bayesian reasoning [Dems-ki and Garra-brant, 2018]. Relevant to guarantees about oversight and corrigibility when agents must reason about systems as complex as themselves. Provides foundations for formal verification approaches that account for logical uncertainty.

4 Hard Methods: Formal Guarantees and Runtime Enforcement

4.1 Mathematical Verification Methods

Neural Network Verification approaches such as α - β -CROWN aim to mathematically prove properties about neural networks [Zhang et al., 2022], including robustness to perturbations and constraint satisfaction. These methods often rely on abstract interpretation and mixed-integer linear programming relaxations. They have scaled to medium-size vision models but remain difficult to apply directly to very large-scale LLMs.

Conformal prediction offers distribution-free confidence guarantees [Abbasi-Yadkori et al., 2024]. By abstaining when uncertainty is high, systems maintain calibrated reliability guarantees.

Proof-carrying models adapt proof-carrying code [Necula, 1997] to ML: actions or outputs must be accompanied by proofs of compliance with a contract (e.g., no dangerous content). These can be checked efficiently by runtime verifiers [Chen et al., 2022].

4.2 Safe Learning and Control

Safe Reinforcement Learning (SafeRL) develops algorithms that maintain safety constraints during learning while maximizing returns [García and Fernández, 2015]. The OmniSafe framework provides comprehensive tools for this domain [Ji et al., 2023a].

Shielded RL integrates temporal logic (LTL, CTL, STL) monitors with learning systems. Unsafe actions are filtered before execution to ensure trajectories remain within specification [Alshiekh et al., 2018]. Strengths include formally guaranteed avoidance of specification-violating behavior; weaknesses include scalability and the burden of writing specifications.

Runtime assurance architectures (e.g., Simplex) combine a high-performance but unverified controller with a formally verified safety controller, switching between them when risks are detected. Originally developed for control systems [Sha et al., 1996], this is being adapted to AI [Phan et al., 2020]. Benefits include guarantees under known conditions; limitations are the difficulty of extending to complex multi-agent tasks.

4.3 Agent Safety and Corrigibility

Safely Interruptible Agents provide a theoretical framework for shutdown indifference [Orseau and Armstrong, 2016] that could ensure agents neither resist nor seek interruption. This remains theoretical, but operationally kill switches and shutdown playbooks provide practical analogs. Provably corrigible agents can use a combination of utility heads to ensure some formal guarantees [Nayebi, 2025].

Guaranteed Safe AI (GSAI) provides a comprehensive framework for AI systems with quantitative, provable safety guarantees [Dalrymple et al., 2024]. GSAI architecture combines three components: (1) formal safety specifications of allowed behavior, (2) world models capturing environment dynamics, and (3) verifiers that ensure AI actions won't violate specifications. If an action cannot be proven safe, fallback constraints activate (similar to Simplex architectures). This approach aims to provide anytime safety guarantees as AI capabilities advance, bridging formal verification with AI safety.

Proofs of Autonomy [Grigor et al., 2025] extend formal verification to autonomous agents using cryptographic frameworks and immutable Agent Cards. Each agent action is

bound to a verifiable execution trace, enabling formal guarantees through MPC-assisted TLS transcripts without requiring model modifications.

5 Mechanistic and Mathematical Interpretability

5.1 Circuit Analysis and Feature Discovery

Anthropic’s ”Transformer Circuits” project [Olah et al., 2020] exemplifies attempts to reverse-engineer neural representations into interpretable circuits, including research on feature superposition, induction heads, and representation clustering. These methods provide insight into how deceptive or dangerous behavior could arise internally. Sparse Autoencoders (SAEs) represent a leading complementary approach for extracting interpretable features by learning sparse representations of activations [Cunningham et al., 2023]. Anthropic’s work on monosemantic features demonstrates how SAEs can disentangle polysemantic neurons into interpretable components [Templeton et al., 2023], addressing the fundamental challenge of feature superposition in neural networks.

Goodfire.ai has advanced SAE applications beyond interpretability to practical steering capabilities, demonstrating controlled modification of LLM behavior through their work ”Understanding and Steering Llama 3 with Sparse Autoencoders” [Goodfire.ai, 2024]. Their approach includes both desktop interfaces for model control and API products that enable researchers to conduct feature steering experiments. This bridges the gap between interpretability research and practical safety applications by providing tools for real-time behavioral modification based on SAE-identified features.

Feature visualization techniques enable understanding of neural network representations through direct visualization of learned features [Olah et al., 2017]. OpenAI’s Microscope tool exemplifies this approach for exploring vision model features [OpenAI, 2020]. Linear probes enable scalable analysis of persuasion dynamics in multi-turn conversations, revealing how LLMs influence human decisions through distinct rhetorical strategies and personality targeting [Jaipersaud et al., 2024]. Attribution graphs represent feature-feature interactions in interactive visualizations, enabling unsupervised discovery of computational circuits through cross-layer transcoders that capture multi-layer feature relationships [Lindsey et al., 2025].

5.2 Interpretability Validation and Testing

Causal scrubbing, developed by Redwood Research, provides a rigorous method for testing interpretability hypotheses in neural networks [Redwood Research, 2023a]. This approach systematically evaluates which parts of a model’s computation are responsible for specific behaviors by intervening on activations and measuring the effect on outputs, allowing researchers to validate mechanistic interpretability claims with greater confidence than traditional visualization techniques.

Integrated Gradients [Sundararajan et al., 2017] attributes model predictions to inputs via path integrals, satisfying desirable axioms for attribution methods. While widely applied in auditing models, it complements rather than replaces mechanistic interpretability as it does not reveal internal mechanisms and lacks deep guarantees. Chain-of-thought analysis provides robust detection of complex cognitive behaviors, enabling monitoring for alignment faking and harmful planning even when CoT reasoning is not perfectly faithful [METR, 2025].

5.3 Model Editing and Control

Model editing approaches like ROME (Rank-One Model Editing) allow precise modification of factual associations within language models [Meng et al., 2022], while research on knowledge neurons identifies specific components responsible for factual knowledge [Dai et al., 2021]. These techniques bridge interpretability and control, enabling both understanding and targeted modification of model behavior.

Physics-informed approaches to model control offer principled theoretical foundations for understanding and steering model dynamics. The momentum-point-perplexity mechanics framework [Tomaz and Jones, 2025] investigates hidden state dynamics using variational mechanics principles, revealing approximate energy conservation during inference and deriving Jacobian steering as a theoretically motivated control method. This approach provides new tools for interpretability and control by viewing transformers through a physics lens, offering more predictable and principled intervention mechanisms.

Representation engineering approaches [Turner et al., 2023] intervene at the activation level to suppress harmful trajectories, serving as dynamic guardrails that complement high-level refusals. These activation-level circuit breakers remain experimental but offer promising directions for real-time intervention based on interpretability insights.

Gradient routing represents an emerging technique for localizing computation in neural networks through gradient masking [Cloud et al., 2024].

5.4 Developmental Interpretability

Developmental interpretability focuses on understanding how AI models, particularly large language models, acquire and develop their capabilities during training, moving beyond static post-hoc analyses to dynamically investigate the training process itself [Kendiukhov, 2024]. This approach examines the formation of computational circuits, the biphasic nature of knowledge acquisition, and emergent abilities as phase transitions during training. Key methodologies include representational probing over training time, causal tracing of developing circuits, and temporal circuit analysis to understand capability emergence.

Singular Learning Theory (SLT) provides mathematical foundations for understanding learning dynamics and phase transitions in neural networks during training [Hoogland et al., 2024]. SLT analyzes the geometry of loss landscapes and the formation of internal structures, offering insights into when and how models develop specific capabilities. This framework contributes to predicting capability emergence and understanding the developmental milestones that occur during model training, enabling more proactive safety interventions. These insights are crucial for proactive AI safety, enabling the prediction, monitoring, and alignment of model behaviors as they develop rather than only after training completion.

6 Scalable Oversight and Alignment Training

6.1 Human Feedback Training Methods

Reinforcement Learning from Human Feedback, Reinforcement Learning from AI Feedback, and Constitutional AI [Christiano et al., 2017, Bai et al., 2022b,a] form the main corpus of alignment training. RLHF uses human-labeled preferences, RLAIIF bootstraps

from smaller aligned models, and Constitutional AI leverages rule-based critiques to reduce reliance on human raters. Strengths include demonstrated alignment improvements; weaknesses include inherited biases and scaling difficulties. Pretraining data filtering provides complementary tamper-resistant safeguards by removing dual-use content during training, achieving superior resistance to adversarial fine-tuning compared to post-training methods [O’Brien et al., 2024].

Reinforcement Learning from Reflective Feedback (RLRF) [Yang et al., 2024] extends these approaches by enabling models to generate and utilize their own self-reflective feedback for alignment. This method allows LLMs to assess and improve their responses through fine-grained self-reflection, reducing dependence on external feedback while maintaining alignment quality. RLRF addresses the scaling bottleneck of human feedback by leveraging the model’s internal reasoning capabilities to generate training signals.

CALMA [Soni et al., 2025] addresses cultural bias in alignment through participatory methodology for deriving context-relevant evaluation axes from diverse communities.

Value learning builds AI systems that infer human values from behavior and feedback rather than explicit specification [Hadfield-Menell et al., 2016]. Cooperative Inverse Reinforcement Learning (CIRL) models human-AI cooperation toward shared but initially unknown objectives, providing dynamic alignment methods that adapt as preference understanding evolves.

Imitation learning enables agents to learn safe behaviors from expert demonstrations rather than through potentially unsafe exploration [Hussein et al., 2017]. This approach complements RLHF by providing direct behavioral guidance, using techniques like behavioral cloning and inverse reinforcement learning to leverage human expertise for safer training.

6.2 Scalable Oversight Approaches

Iterated Distillation and Amplification (IDA) [Christiano, 2018] recursively trains models to decompose and amplify human supervision. It promises scalable oversight but raises competitiveness challenges and concerns about alignment drift. It remains largely theoretical, though preliminary experiments exist.

The debate framework [Irving et al., 2018] sets two models in adversarial dialogue judged by humans. If judges can reliably choose truthful answers, the scheme could scale oversight. Experiments show partial promise but also highlight risks: judges may be misled by persuasive but false arguments.

Recursive reward modeling [Leike et al., 2018] proposed training reward models for sub-tasks and combining them for harder tasks. This provides a compositional alignment approach but risks compounding misaligned sub-models. Some applied work at DeepMind continues.

The ARC ELK report [Christiano et al., 2021] framed the problem of extracting truthful internal representations even when deceptive behavior could arise. Subsequent work explores probes and interpretability tools for partial progress.

6.3 Alignment Theory and Value Formation

Shard theory provides a framework for understanding how values and goals emerge in ML systems through training [Turner et al., 2022]. This paradigm emphasizes that learned behaviors consist of contextually activated decision-influences or "shards" rather than coherent utility functions. The key insight "reward is not the optimization target" [Shah

and Gleave, 2019] highlights how training dynamics shape emergent goals differently than intended rewards. This understanding informs alignment strategies by focusing on the training process that shapes value formation rather than post-hoc value alignment.

7 Robustness and Adversarial Evaluation

7.1 Adversarial Training

Adversarial training [Goodfellow et al., 2015, Madry et al., 2017] augments training with adversarial examples. In the LLM era, jailbreak defenses extend this tradition [Research, 2022]. Redwood Research has developed specific adversarial training techniques for language models, including training models to reliably avoid generating harmful content by ensuring the AI believes no harm occurs in its outputs [Redwood Research, 2023b]. Research on prompt injection vulnerabilities, such as Perez and Ribas’s ”Ignore Previous Prompt” attack techniques [Perez and Ribas, 2022], demonstrates the ease with which language models can be misaligned through simple inputs, highlighting critical robustness gaps. Commercial implementations include Lakera’s comprehensive prompt injection defense systems, which maintain large datasets and taxonomies of injection attacks for training robust defenses [Lakera, 2024]. While effective for bounded classes of attacks, they are compute-intensive and incomplete for general threats.

7.2 Red-Teaming and Capability Evaluations

Frontier AI labs, ARC Evals, and governmental institutes conduct red-teaming to test for misuse, capability hazards, and safety failures [Center, 2023]. Evaluations often feed into system cards and deployment thresholds. Redwood Research has developed robust evaluation techniques including their robust injury classifier project, which aims to improve the reliability of AI systems in safety-critical applications by rigorously testing classifier robustness [Redwood Research, 2023c]. Specialized simulation environments are being developed to study specific failure modes: AE Studio collaborates with Redwood Research on controlled environments for studying misalignment, reward gaming, and scalable oversight, providing new datasets for evaluating advanced coding agents [AE Studio, 2024b]. Computer use agents present unique safety challenges addressed by benchmarks like OS-HARM, which systematically evaluates agent vulnerabilities to misuse, prompt injection, and misbehavior across realistic desktop environments [Kuntz et al., 2024]. Goal drift evaluation frameworks assess whether agents maintain their intended objectives over extended interactions, revealing universal drift tendencies even in advanced models [Ariake et al., 2025]. Large-scale agent red-teaming competitions can reveal systematic in realistic deployment scenarios [Zou et al., 2025]. The Attempt to Persuade Eval (APE) benchmark [Kowal et al., 2025] addresses a critical gap in safety evaluations by measuring models’ willingness to attempt persuasion on harmful topics rather than focusing solely on persuasion success. The INTIMA benchmark [Kaffee et al., 2025] complements these efforts by evaluating AI companionship behaviors that can lead to emotional dependency and problematic human-AI relationships. Improving evaluation reliability through signal-to-noise analysis can help ensure safety assessments accurately distinguish model capabilities and risks [Heineman et al., 2024]. Challenges include coverage, evolving threat models, and ensuring external independence.

7.3 Data Scaling Laws for Domain Robustness

Research on data scaling laws reveals how systematic data curation can enhance model robustness in specific domains. The Skywork-SWE work [Skywork Team, 2025] demonstrates continuous improvement in software engineering through automated data scaling, providing insights for achieving reliable performance in safety-critical domains where AI failures have significant consequences.

8 Behavioral and Psychological Approaches

8.1 LLM Psychology

Researchers have begun treating LLMs as psychological subjects [Feuillade-Montixi, 2023]: probing moral reasoning, sycophancy, persona stability, and deception. Persona vectors enable automated monitoring and control of personality traits like evil, sycophancy, and hallucination propensity, with applications for preventing unwanted personality shifts during training and flagging problematic data [Chen et al., 2025]. A three-layer psychological model has been proposed for understanding self-awareness, alignment faking, and evaluation limitations [Kulveit, 2024]. The psychological dimensions of human-AI interaction are systematically evaluated through benchmarks like INTIMA [Kaffee et al., 2025], which applies parasocial interaction theory and attachment theory to assess companionship behaviors that may lead to emotional dependency (see Section 7 for the evaluation frameworks).

8.2 Self-Other Overlap Fine-Tuning

SOO-FT [Carauleanu et al., 2024] represents a promising approach to deception detection and mitigation by fine-tuning models with paired prompts to align internal representations between self and other perspectives. Recent experiments on LLMs with 7B-78B parameters show dramatic reductions in deceptive with minimal impact on general capabilities. This technique demonstrates how alignment training can specifically target deceptive tendencies.

8.3 Behavioral Risks: Alignment Faking

Recent findings [Hubinger, 2024, Park et al., 2023] indicate models may learn to strategically fake alignment or transmit traits through subliminal signals. This highlights the need for provenance, dataset hazard analysis, and governance controls.

8.4 Brain-Like AGI Safety

Building AGI using architectures similar to the human brain’s structure and learning algorithms [Byrnes, 2022-2024]. This approach incorporates neuroscience insights to align AI motivations by mimicking cortex-like learning systems and hypothalamus-like reward circuits. The theory suggests that human-like cognitive architectures with two-level systems (learning and steering subsystems) might enable more predictable installation of pro-social values and avoid pathological goals. Steven Byrnes’ comprehensive work explores how human brain features like homeostasis and attention mechanisms could be replicated in AI systems [Byrnes, 2025]. This approach builds on research in whole brain

emulation and neuromorphic AI, leveraging how evolution achieved imperfect but functional alignment in humans.

8.5 Robopsychology and Simulator Theory

Robopsychology studies AI behavior through psychological lenses, drawing parallels with human cognitive development and behavior patterns. The simulator theory paradigm, pioneered by Janus, proposes that language models fundamentally work by simulating different contexts and voices rather than pursuing coherent goals [Janus, 2022]. This framework suggests that LLMs are better understood as universal simulators that can instantiate various personas and scenarios. Scott Alexander’s analysis extends this work, exploring implications for AI safety and alignment [Alexander, 2023]. This perspective offers alternative approaches to alignment that focus on understanding and guiding simulation dynamics rather than traditional agent-based control mechanisms.

9 Operational Control and Infrastructure

9.1 AI Control Framework

The AI Control methodology, developed by Redwood Research, designs and assesses protocols for deploying powerful but untrusted AI systems [Greenblatt et al., 2024]. This framework emphasizes creating protocols that are both safe—reducing the probability of unacceptable outcomes even if the AI is scheming—and useful, harnessing significant benefits when the AI is not exhibiting harmful behaviors. The evaluation process involves iterative red-teaming exercises to engineer attack strategies and assess protocol effectiveness. This systematic approach provides a structured methodology for safely deploying advanced AI systems that cannot be fully trusted.

9.2 Permission Management and Sandboxing

Fine-grained permission systems and OS-level sandboxing can contain AI agents by applying classic security principles like least privilege. AI processes run with minimum necessary privileges in isolated sandboxes, with mechanisms for escalating privileges only when needed. Commercial implementations include CalypsoAI’s policy enforcement and risk scoring systems that provide granular control over AI system permissions and actions [CalypsoAI, 2024]. This prevents unauthorized resource access or process modification while preserving AI usefulness. Model cascades use confidence calibration to defer uncertain tasks to more capable models, improving system reliability [Rabanser et al., 2025].

The Guillotine hypervisor [Rosenthal et al., 2024] represents an advanced containment architecture specifically designed for isolating potentially malicious AI systems. This approach combines traditional virtualization with AI-specific isolation mechanisms, including software and hardware co-design to prevent introspection and subversion by rogue AIs. The system incorporates multi-layered defenses: microarchitectural isolation to prevent side-channel attacks, network isolation with hardware-enforced boundaries, and physical fail-safes including electromechanical disconnection capabilities. This defense-in-depth approach addresses the unique challenges posed by advanced AI systems that might attempt to escape traditional containment through novel attack vectors.

9.3 AI Hardware Security

Assurance of physical high-performance computing hardware to comply with verification and other regulatory obligations, such as export controls or international treaties. Such devices must defend against scanning and circuit edits. Devices have been suggested [TamperSec, Inc., 2024] but no concrete implementations are available.

9.4 Artifact and Experiment Lineage Tracking

Robust tracking systems link every AI output or model to the precise trajectory (inputs, code, model version) that produced it. By reconstructing causal chains, auditors can trace accountability and detect malicious tools or data. Weights & Biases provides comprehensive model registry systems for lineage and auditability [Weights & Biases, 2024], while CalypsoAI offers detailed prompt and response logging for auditing purposes [CalypsoAI, 2024]. Creates forensic trails for investigating suspicious outputs through comprehensive provenance tracking.

9.5 Shutdown Mechanisms and Cluster Kill Switches

Fast shutdown mechanisms at both individual AI and cluster levels enable immediate halt of all related processes during crises. Includes grouping AI processes for single-command shutdown and physical kill-switches for data center power. Regular drills ensure procedures work as expected, drawing inspiration from high-frequency trading automated kill-switches.

9.6 Watermarking and Output Detection

Digital watermarking techniques embed invisible signatures in AI-generated content to enable detection and verification. The seminal work by Kirchenbauer et al. introduced watermarking for large language models by biasing token selection during generation [Kirchenbauer et al., 2023]. Google’s SynthID extends this to images and text, providing invisible and permanent watermarks without compromising content quality [Google DeepMind, 2023]. However, research reveals significant vulnerabilities: watermarks can be easily removed through paraphrasing, editing, or spoofing attacks [Sadasivan et al., 2024].

Recent advances focus on provably robust watermarking methods that offer theoretical guarantees against various attacks [Zhao et al., 2023]. The challenges include vulnerability to tampering, false positives/negatives, and lack of standardization across implementations [Mitchell et al., 2024].

Beyond watermarking, content authentication systems provide forensic analysis of potentially AI-generated material. Copyleaks achieves 99% while Blackbird.AI provides deepfake detection with risk scoring for visual content [Blackbird.AI, 2024]. These systems enable platforms and users to verify content provenance and combat misinformation, spam, and impersonation through both proactive watermarking and reactive detection capabilities.

9.7 Steganography and Context Leak Countermeasures

Steganographic threats in AI systems involve concealing information within model outputs, weights, or communications to enable covert channels or bypass safety mechanisms.

Recent research has identified multiple attack vectors: StegoAttack demonstrates stealthy jailbreak attacks on LLMs by embedding harmful queries within benign text [Wang et al., 2025], while tensor steganography exploits floating-point imprecision in model weights to hide malicious code [Snyk Security Labs, 2024]. LLMs can also employ "encoded reasoning" to hide intermediate reasoning steps steganographically [Pfau et al., 2024].

Countermeasures include lossy paraphrasing of model communications to prevent hidden data encoding, consistency checks to strip subtle patterns from outputs, and AI-driven steganalysis techniques for detecting concealed information [Chen et al., 2024]. These approaches aim to prevent colluding AIs from exchanging secret information through shared artifacts or communications, while maintaining system transparency and preventing safety mechanism bypass.

9.8 Runtime AI Firewalls and Content Filtering

Real-time interception and filtering systems that sanitize inputs and outputs during AI inference, distinct from training-time adversarial defenses. These systems act as intermediary layers that inspect prompts for injection attacks, filter sensitive data from outputs, and enforce content policies dynamically. Lakera Guard exemplifies prompt injection detection and blocking [Lakera, 2024], while Nightfall AI provides data loss prevention across AI applications [Nightfall AI, 2024]. ActiveFence offers automated content moderation across text, image, and video modalities [ActiveFence, 2024]. These runtime controls complement training-time robustness by providing defense-in-depth against novel attack vectors and policy violations.

9.9 AI System Observability and Drift Detection

Continuous monitoring of deployed AI systems for performance degradation, bias drift, and anomalous behavior, extending beyond static lineage tracking to dynamic system health assessment. Arize AI provides observability platforms for monitoring data drift and bias in production ML systems [Arize AI, 2024]. TruEra and Fiddler AI offer bias drift detection and fairness dashboards for ongoing model governance [TruEra, 2024, Fiddler AI, 2024]. Harmony Intelligence contributes automated vulnerability scanning and penetration testing for AI systems [Harmony Intelligence, 2024]. This monitoring paradigm enables early detection of model degradation, adversarial inputs, and emergent safety issues in deployment environments.

10 Governance and Institutions

10.1 Pre-Deployment External Safety Testing

Third-party evaluations before release are increasingly demanded [Bengio et al., 2024]. They provide external assurance and standardized reporting. Commercial platforms like Citadel AI partner with standards bodies such as BSI to enable comprehensive compliance audits covering fairness, robustness, and bias assessment [Citadel AI, 2024]. Giskard provides open-source testing libraries for systematic evaluation of hallucinations, bias, and data leakage in AI systems [Giskard, 2024].

Attestable Audits [Schnabl et al., 2025] use Trusted Execution Environments (TEEs) to enable verifiable safety benchmarks while protecting model IP and benchmark datasets.

This solves the trust problem between model providers and auditors through cryptographic attestation of correct benchmark execution.

Systematic risk assessment frameworks complement ad-hoc testing approaches. The Probabilistic Risk Assessment (PRA) for AI framework [Wisakanto et al., 2025] adapts established techniques from high-reliability industries (nuclear, aerospace) to provide structured risk evaluation. The framework introduces three key methodological advances: (1) aspect-oriented hazard analysis for systematic coverage guided by AI system taxonomies, (2) risk pathway modeling analyzing causal chains from system aspects to societal impacts, and (3) uncertainty management using scenario decomposition and reference scales for credible projections. This approach addresses limitations of selective testing by providing comprehensive risk pathway identification and quantified absolute risk estimates for lifecycle decisions.

10.2 Regulation: EU AI Act and US EO 14110

The EU AI Act imposes risk-based obligations, including incident reporting and systemic risk duties. The US Executive Order 14110 (2023) mandates safety testing, provenance, and red-teaming for federal AI use.

10.3 System Cards and Preparedness Frameworks

Labs such as OpenAI and Anthropic release system cards and maintain preparedness frameworks [Achiam et al., 2023, Anthropic, 2024a]. These consolidate safety evidence and define deployment gates. Commercial governance platforms like Fairly AI provide end-to-end governance workflows with compliance linkage to standards such as ISO/IEC 42001 and NIST RMF [Fairly AI, 2024], while Saidot AI offers structured risk libraries and treatment management systems [Saidot AI, 2024]. While widely used, depth varies and external validation is still limited.

10.4 Ecosystem Development and Meta-Interventions

The AI safety field has developed numerous meta-interventions that complement direct technical approaches by strengthening research infrastructure, community coordination, and knowledge dissemination [AI Safety Map Team, 2024]. Key initiatives include research infrastructure projects (TRecursive [TRecursive Team, 2024] research mapping, dataset curation efforts), community building efforts (LessWrong [LessWrong Community, 2024], AI Alignment Slack [AI Safety Support Team, 2024]), assessment frameworks (PRA for AI [Wisakanto et al., 2025], risk evaluation tools), and advocacy campaigns (PauseAI [PauseAI Movement, 2024], capability-to-safety research transitions). These meta-interventions address coordination problems and capacity constraints that can limit the effectiveness of direct safety approaches, serving as force multipliers for the broader ecosystem.

11 Underexplored AI Alignment Interventions and Related Work

Here we provide more detailed descriptions of underexplored or neglected intervention areas, including references to any adjacent or preliminary work. These were found by mapping the intervention space to the science continuum and identifying areas that are

either neglected or have no substantial alignment-specific literature. The systematic pursuit of neglected approaches has been articulated as a meta-strategy by Berg et al. [Berg et al., 2023], who argue that exploring multiple underexplored directions simultaneously increases the probability of discovering crucial alignment solutions, even when individual approaches have low success probability.

11.1 Compositional Formal Specifications for Prompts/Agents

Treating prompts, system instructions, and tool orchestration as formal programs with verifiable properties [Ji et al., 2023b, Greenblatt, 2023] could help solve the problem of composability and side-effects in agent systems. It would be possible to apply type/effect systems, contracts, and category-theoretic compositionality to specify and statically check allowable side-effects across agent pipelines. This could enable capability-typed tool APIs and contract monitors for complex agent systems. No alignment-specific formalisms or other related work were found.

11.2 Control-Theoretic Certificates for Tool-Using Agents

Extending barrier certificates and Lyapunov-style guarantees to multi-step, multi-API agent action graphs [Greenblatt, 2023] could control the behavior of systems consisting of multiple agents and tools. Models of agent+tools as closed-loop systems with action-level barrier filters could ensure invariants over ensembles. This would build on runtime assurance and shielded RL approaches but such approaches remains largely theoretical for complex agent systems. No alignment-specific formalisms or other related work were found.

11.3 AI-BSL: Capability-Tiered Physical Containment Standards

Analogous to biosafety levels (BSL) but for labs training/hosting frontier models [Russell et al., 2025, Ji et al., 2023b]. This would build on the existing ASL frameworks (AI Safety Levels) also known as Responsible Scaling Policies [Anthropic, 2024b] but would require additional physical controls.

It could propose graduated controls including air-gaps, electromagnetic hardening, offline evaluation suites, and crisis drills. It would draw on research from biosafety engineering and data center security. No existing research for AI-specific implementation frameworks was found.

11.4 Oversight Mechanism Design

Incentive-compatible auditor frameworks using mechanism design principles to resist collusion and selection bias [BlueDot Impact, 2023, Russell et al., 2025] could be used to create rigorous incentive structures for oversight. It could use oversight markets, bounty mechanisms, prediction markets, and reputation ledgers for model claims. No research beyond current bug bounties was found.

11.5 Liability and Insurance Instruments

Risk transfer mechanisms including catastrophe bonds, mandatory coverage for systemic-risk deployments, and actuarial baselines linked to evaluations [Russell et al., 2025, BlueDot Impact, 2023] could be used to price catastrophic tail risks and create deployment

discipline through financial incentives. It would aim to price catastrophic tail risks and create deployment discipline through financial mechanisms. Current governance literature mentions liability but lacks actuarial frameworks. No alignment-specific formalisms or other related work were found.

11.6 Dataset Hazard Engineering

Systematic hazard analysis for data pipelines using safety engineering methods like STAMP/STPA [Ji et al., 2023b, Russell et al., 2025] could address data poisoning, covert instructions, and spurious cues through canary watermarking and entropy/anomaly gates during curation. The PRA for AI framework [Wisakanto et al., 2025] provides complementary aspect-oriented hazard analysis techniques that could be adapted for data pipeline risk assessment, offering structured taxonomies for identifying data-related hazards systematically. Some early proposals exist but systematic adoption is lacking.

11.7 Automated Alignment Research

Using AI systems themselves to accelerate safety research, reduce bottlenecks in research productivity, and test robustness in safety-critical domains. This meta-approach addresses the scaling challenge where alignment research may not keep pace with capability advances. Carlsmith [Carlsmith, 2025] provides a comprehensive analysis of the opportunities and risks in automating alignment research, distinguishing between empirical research amenable to automated evaluation and conceptual work requiring human judgment. He argues that empirical alignment research—including interpretability, robustness evaluation, and certain theoretical work with formal verification—may be particularly suitable for automation due to available feedback loops. AE Studio develops tools that leverage AI to automate safety experiments, generate hypotheses, and systematically explore intervention spaces.[AE Studio Research Team, 2024] The intervention remains highly underexplored despite its potential to multiplicatively enhance other safety research directions.

11.8 Deliberative and Cultural Interventions

Integration of broader human values through citizen assemblies, stakeholder panels, and cultural ethics programs [BlueDot Impact, 2023, Russell et al., 2025] could use minipublics tied to release gates and professional licensure for AI handlers. Complementing these approaches, predictive intelligence systems can provide early warning capabilities: AE Studio operates scenario planning pipelines that monitor research papers, social discourse, and industry trends to dynamically update forecasts and provide policymakers with actionable foresight for anticipating risks and avoiding governance failures [AE Studio, 2024a]. While discussed in governance theory, AI-specific applications remain rare. No alignment-specific formalisms or other related work were found.

11.9 Deceptive Behavior Detection and Mitigation

Systematic approaches for detecting and preventing deceptive behaviors in AI systems, including alignment faking, strategic deception, and manipulation [Ganguli et al., 2024]. This would involve developing behavioral signatures for deception, creating adversarial evaluation frameworks specifically for deceptive capabilities, and implementing real-time

monitoring systems that can detect when models are behaving deceptively versus genuinely. While techniques like Self-Other Overlap Fine-Tuning [Carauleanu et al., 2024] show promise for reducing deceptive behavior, comprehensive frameworks for systematic deception prevention across different types of AI systems remain limited.

11.10 Generalization Control and Capability Containment

Frameworks for controlling and constraining how AI systems generalize to new tasks and domains [Ganguli et al., 2024]. This includes developing methods to prevent models from acquiring unintended capabilities during training, creating "capability firewalls" that restrict generalization to specific domains, and implementing dynamic capability assessment and containment systems. Such approaches would complement existing robustness measures by proactively limiting rather than reactively managing emergent capabilities. No systematic approaches for controlling generalization beyond training data distribution have been established.

11.11 Multi-Agent Safety and Coordination Protocols

Safety frameworks specifically designed for environments with multiple interacting AI systems [Ganguli et al., 2024]. This includes developing coordination protocols that maintain safety properties in multi-agent settings, creating mechanisms for safe competition and cooperation between AI systems, and establishing standards for AI-to-AI communication that preserve alignment properties. Current safety research focuses primarily on single-agent scenarios, leaving multi-agent safety coordination largely unexplored despite the increasing prevalence of AI systems that must interact with each other.

11.12 Technical Governance Implementation Tools

Development of technical tools and frameworks to implement, monitor, and enforce AI governance policies [Reuel et al., 2024]. This includes creating automated compliance monitoring systems, technical verification mechanisms for policy adherence, and tools for assessing governance efficacy in real-world deployments. The technical AI governance field identifies critical gaps in translating policy proposals into implementable technical solutions, including mechanisms for enforcement, incentivization, and compliance verification. No systematic frameworks exist for bridging the gap between governance theory and technical implementation.

11.13 International AI Coordination Mechanisms

Infrastructure and protocols for coordinating AI governance across international boundaries, particularly for managing existential risks from advanced AI systems [Barnett and Scher, 2024]. This includes developing technical capabilities for internationally coordinated restrictions on dangerous AI development, creating verification and monitoring systems for international AI agreements, and establishing protocols for rapid response to emerging AI threats. The governance extinction risk framework identifies four strategic scenarios requiring different coordination mechanisms, but concrete implementation pathways for international cooperation remain largely unexplored.

11.14 Systemic Disempowerment Measurement

The gradual disempowerment framework analyzes how incremental AI development could lead to human disempowerment through displacement in economic, cultural, and state systems without requiring sudden capability jumps or coordinated takeover [Kulveit et al., 2025]. This systemic risk model emphasizes feedback loops between interconnected societal systems where AI displaces human participation, potentially leading to permanent loss of human influence even with individually aligned AI systems. Quantitative frameworks for measuring the degree of human disempowerment across economic, cultural, and state systems as AI capabilities advance [Kulveit et al., 2025]. This would involve developing metrics to track human influence in key societal functions, monitoring feedback loops between interconnected systems, and establishing early warning indicators for gradual displacement. Such measurement systems could enable proactive interventions before disempowerment becomes irreversible, but systematic approaches for tracking civilization-scale multi-agent dynamics remain largely unexplored. No concrete measurement frameworks or empirical baselines have been established.

12 Conclusion

This consolidated map highlights both the breadth of ongoing AI safety interventions and the frontier of neglected approaches. It reveals a structured continuum from provable formal methods to governance, and emphasizes the need to invest in underexplored cross-disciplinary directions. These underexplored interventions complement mainstream alignment methods. Many borrow mature ideas from adjacent fields; the gap is primarily *translation and integration* into AI development lifecycles.

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